NOPT061 Tentative schedule

- **Lecture 1 - Introduction, Heuristics, Metaheuristics**
  - Combinatorial optimization
    - Examples: Traveling Salesman Problem (TSP), Knapsack
    - Exact method, heuristic, metaheuristic, hybrid metaheuristic
  - Basic heuristic methods
    - Constructive heuristics (example: greedy)
    - Local search (LS)
      - extension: Variable neighborhood descent
  - Metaheuristics
    - Greedy randomized adaptive search procedure (GRASP)
    - Iterated greedy algorithms
    - Iterated local search
    - Simulated annealing
    - Tabu search
    - Variable neighborhood search (VNS)
    - Ant colony optimization (ACO)
  - **Resources:** [powerful] Introduction, 1.1-1.2 (skip 1.2.8)
- **Lecture 2 - Evolutionary and Memetic Algorithms, Exact Methods**
  - Evolutionary and memetic algorithms
    - Evolutionary algorithms
    - Memetic algorithms
    - Lamarckian vs. Baldwinian learning
  - Exact methods
    - Tree search methods (Branch and Bound, example: Knapsack)
    - Dynamic programming (example: Knapsack)
    - Mixed integer programming (MIP) (review of LP, if necessary)
    - Constraint programming (CP)
  - **Resources:** [powerful] Subsection 1.2.8, Section 1.7; [handbook] Chapter 9 (An Accelerated Introduction to Memetic Algorithms); [comparing-memetic] Section 2 (and the rest)
- **Lecture 3 - Incomplete Solutions and Decoders**
  - Incomplete/indirect solution representations
  - Decoder-based metaheuristics
  - Application: Generalized Minimum Spanning Tree

    * Initial solutions
    * Node set based VNS
    * Global edge set based VNS
    * Combined VNS
  - **Resources:** [powerful] Chapter 2

- **Lecture 4 - Problem Instance Reduction**
  - Construct, Merge, Solve & Adapt Algorithm (CSMA)
  - Application: Minimum Common String Partition
    * Probabilistic solution generation
    * Solving reduced sub-instances
  - **Resources:** [powerful] Chapter 3

- **Lecture 5 - Large Neighborhood Search**
  - MIP-based Large neighborhood search
  - Application: Minimum Weight Dominating Set
    * initial solution (greedy heuristic)
    * partial destruction of solutions
  - Application: Generalized Minimum Spanning Tree
    * global subtree optimization neighborhood
  - **Resources:** [powerful] Chapter 4

- **Lecture 6 - Parallel Non-independent Construction**
  - General idea
  - Beam Search
  - Beam-ACO: Beam search + Ant Colony Optimization
  - Application: Multidimensional Knapsack (MK)
    * greedy heuristic
    * beam search for MK
    * pure-ACO approach to MK
    * Beam-ACO for MK
  - **Resources:** [powerful] Chapter 5

- **Lecture 7 - Construction and Constraint Programming**
  - Hybrids of constructive metaheuristics and constraint programming
  - Ant-Colony optimization + constraint programming integration
  - Application: Machine Scheduling with Sequence-dependent Setup Times

- **Resources:** [emerging] Paper 6

- **Lecture 8 - Complete solution archives**
  - General idea
  - Trie-based archive
  - Application: Generalized Minimum Spanning Tree (GMST)
    - Archive-Enhanced Evolutionary Algorithm for GMST
  - **Resources:** [powerful] Chapter 6, Sections 6.1 & 6.3

- **Lecture 9 - Hyper-heuristics**
  - Introduction
  - Hyper-heuristics for
    - Boolean Satisfiability
    - Timetabling
    - Packing
  - **Resources:** [search-methodologies] Chapter 20

- **Lecture 10 - Recent trends and applications**
  - Fixed Set Search Applied to the Traveling Salesman Problem
  - Ant Colony Optimization with Policy Gradients and Replay (intro)
  - **Resources:** [fixed-set-search], [policy-gradients]

- **Lecture 11 - Recent trends and applications**
  - Ant Colony Optimization with Policy Gradients and Replay (continued)
  - Letting a Large Neighborhood Search for an Electric Dial-A-Ride Problem Fly: On-The-Fly Charging Station Insertion
  - **Resources:** [policy-gradients], [on-the-fly-insertion]